

|               |                                                      |
|---------------|------------------------------------------------------|
| <b>Ex.No:</b> | <b>Implement Linux Privilege Escalation Checker.</b> |
| <b>Date:</b>  |                                                      |

**AIM:**

**ALGORITHM:**

231401102

## CODE:

```
import subprocess
def run(command):
    try:
        result =
            subprocess.run( com
            mand, shell=True,
            text=True,
            capture_output=True
        )
        return result.stdout.strip()
    except Exception:
        return "Unable to execute command"
print("Linux Privilege Escalation Checker")
print("-" * 40)
print("\n1. Current User")
print(run("whoami"))
print(run("id"))
print("\n2. Sudo Permissions")
sudo = run("sudo -n -l 2>/dev/null")
if sudo:
    print(sudo)
else:
    print("No passwordless sudo permission found.")
print("\n3. SUID Files")
suid = run(
    "find /usr/bin /usr/sbin /bin /sbin "
    "-perm -4000 -type f 2>/dev/null"
)
if suid:
    print(suid)
else:
    print("No SUID files found.")
print("\n4. World-Writable Files")
writable = run(
    "find /etc /usr /opt "
    "-type f -perm -0002 2>/dev/null | head -20"
)
if writable:
    print(writable)
else:
    print("No world-writable files found.")
print("\n5. Sensitive File Permissions")
```

```
print(run("ls -l /etc/passwd /etc/shadow 2>/dev/null"))
print("\n6. Writable Cron Files")
cron = run(
    "find /etc/cron* -type f -writable 2>/dev/null"
)
if cron:
    print(cron)
else:
    print("No writable cron files found.")
print("\nScan Completed Successfully.")
```

## OUTPUT:

Linux Privilege Escalation Checker

### 1. Current User

root  
uid=0(root) gid=0(root) groups=0(root)

### 2. Sudo Permissions

Matching Defaults entries for root on e67c19c7c285:

env\_reset, mail\_badpass, secure\_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/

User root may run the following commands on e67c19c7c285:

(ALL : ALL) ALL

### 3. SUID Files

/usr/bin/passwd  
/usr/bin/umount  
/usr/bin/mount  
/usr/bin/gpasswd  
/usr/bin/su  
/usr/bin/newgrp  
/usr/bin/chfn  
/usr/bin/chsh  
/usr/bin/sudo  
/usr/bin/fusermount  
/usr/bin/pkexec

#### 4. World-Writable Files

No world-writable files found.

#### 5. Sensitive File Permissions

-rw-r--r-- 1 root root 1185 Aug 20 13:15 /etc/passwd

-rw-r----- 1 root shadow 633 Aug 20 13:15 /etc/shadow

#### 6. Writable Cron Files

/etc/cron.d/e2scrub\_all

/etc/cron.daily/dpkg

/etc/cron.daily/apt-compat

/etc/cron.daily/man-db

/etc/cron.weekly/man-db

Scan Completed Successfully.

**RESULT:**